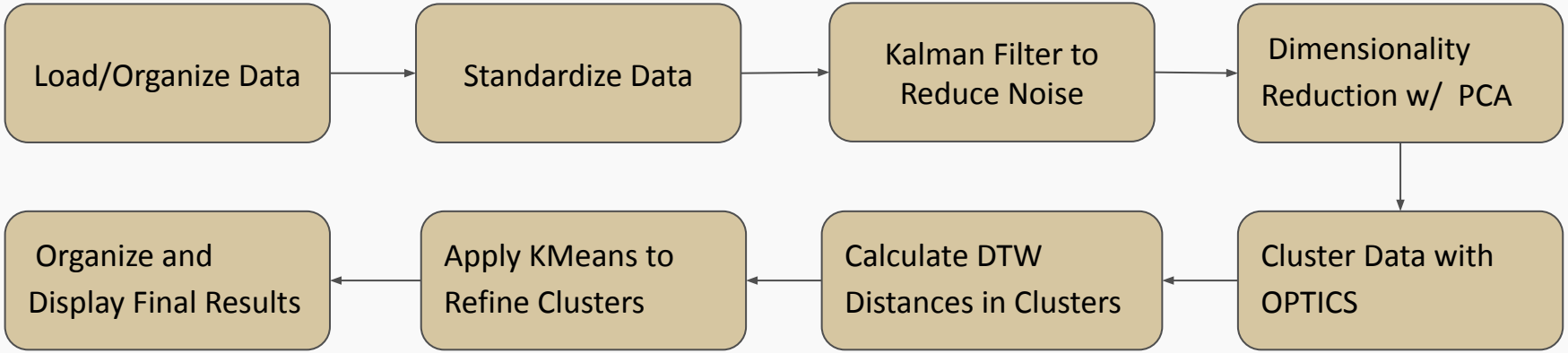
FQE

Financial Quants and Engineers Club

Pair Trading Algorithm

11/18/2024

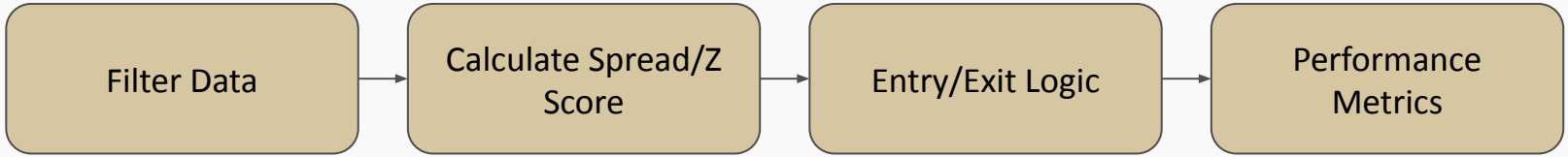
Overview


Load/Organize Data
Standardize Data
Kalman Filter to Reduce Noise
Dimensionality Reduction w/ PCA
Cluster Data with OPTICS
Calculate DTW Distances in Clusters
Apply KMeans to Refine Clusters
Organize and Display Final Results

Add Firm Data

```
firm_data = pd.read_csv("stock_performance/firm_level.csv")
firm_data.rename(columns={'permno': 'PERMNO'}, inplace=True)
firm_data['public_date'] = pd.to_datetime(firm_data['public_date'])

firm_data['divyield'] = firm_data['divyield'].str.rstrip('%').astype('float') / 100.0
numeric_cols = ['pe_exi', 'dpr', 'roe', 'GProf', 'divyield']
```

Backtest


Filter Data
Calculate Spread/Z Score
Entry/Exit Logic
Performance Metrics

Trading Strategies

Kalman Filter State Transition Equation:

$$\mathbf{x}k = \mathbf{F}\mathbf{x}k - 1 + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k$$

xk = State vector at time k .

F = State transition matrix.

B = Control input matrix.

Uk = Control vector.

Wk = Process noise (assumed to be Gaussian).

Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

a(i) : Mean intra-cluster distance (average distance between i and all other points in the same cluster).

b(i) : Mean nearest-cluster distance (average distance between i and all points in the nearest neighboring cluster).

Backtesting Ideas

Rolling Mean/STD:

```
# Calculate rolling mean and std of the spread
mean_spread = spread.rolling(window=30).mean()
std_spread = spread.rolling(window=30).std()
z_score = (spread - mean_spread) / std_spread
```

Limit Date

```
pivot_data = pivot_data.loc['2023-01-01':'2023-12-29']
```

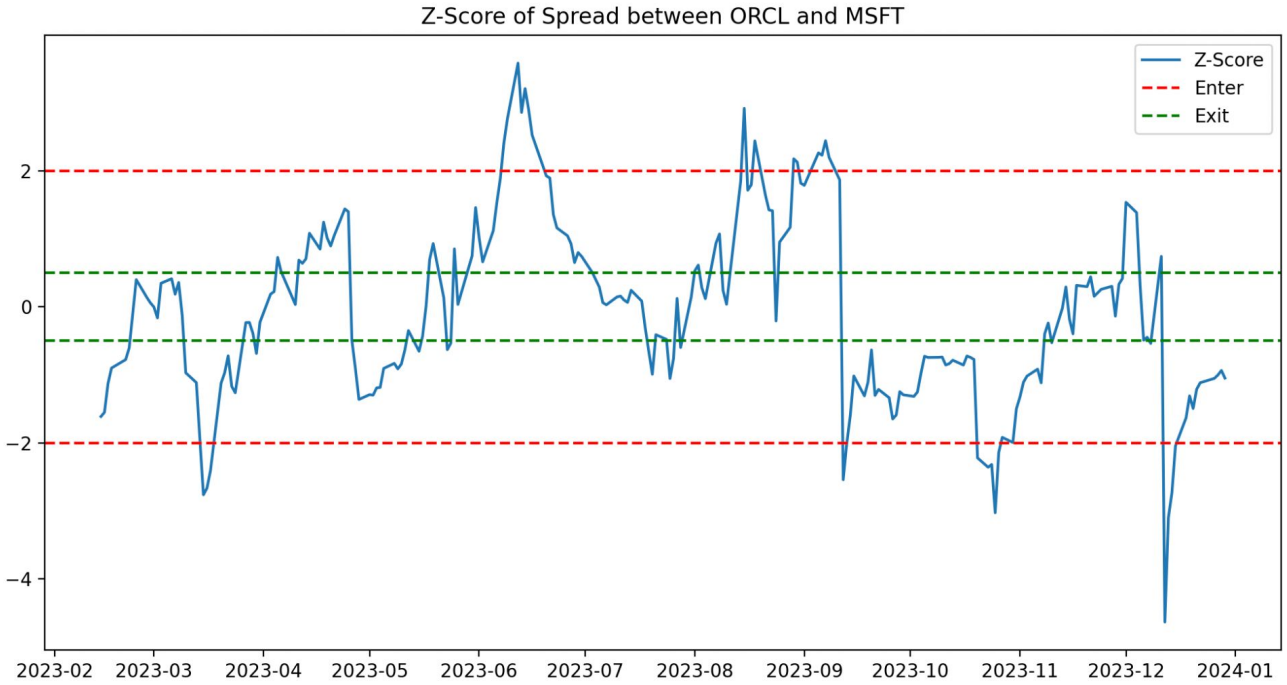
Entry Signal when both stocks exceed two STDs Z-score

```
# Entry signal
if position == 0:
    if z_score.iloc[i] > z_entry:
        # Short spread: Sell stock1, buy stock2 * beta
        position = -1
        entry_price = (price1, price2)
        trade_capital = capital * trade_size_fraction
        shares_stock1 = trade_capital / (price1 + beta * price2)
        shares_stock2 = beta * shares_stock1
        entry_shares_stock1 = -shares_stock1
        entry_shares_stock2 = shares_stock2
        trades.append({
            'date': date,
            'position': position,
            'entry_price': entry_price,
            'entry_shares_stock1': entry_shares_stock1,
            'entry_shares_stock2': entry_shares_stock2
        })
```

Exit Signal when both stocks are within .5 STDs Z-score

```
# Exit signal
elif position != 0 and abs(z_score.iloc[i]) < z_exit:
    exit_price = (price1, price2)
    pnl_stock1 = (exit_price[0] - entry_price[0]) * entry_shares_stock1
    pnl_stock2 = (exit_price[1] - entry_price[1]) * entry_shares_stock2
    cost_stock1 = abs(entry_shares_stock1) * (entry_price[0] + exit_price[0]) * transaction_cost
    cost_stock2 = abs(entry_shares_stock2) * (entry_price[1] + exit_price[1]) * transaction_cost
    total_cost = cost_stock1 + cost_stock2
    pnl = pnl_stock1 + pnl_stock2 - total_cost
    portfolio.append(pnl)
```

One of the closest pairs with Firm level Data



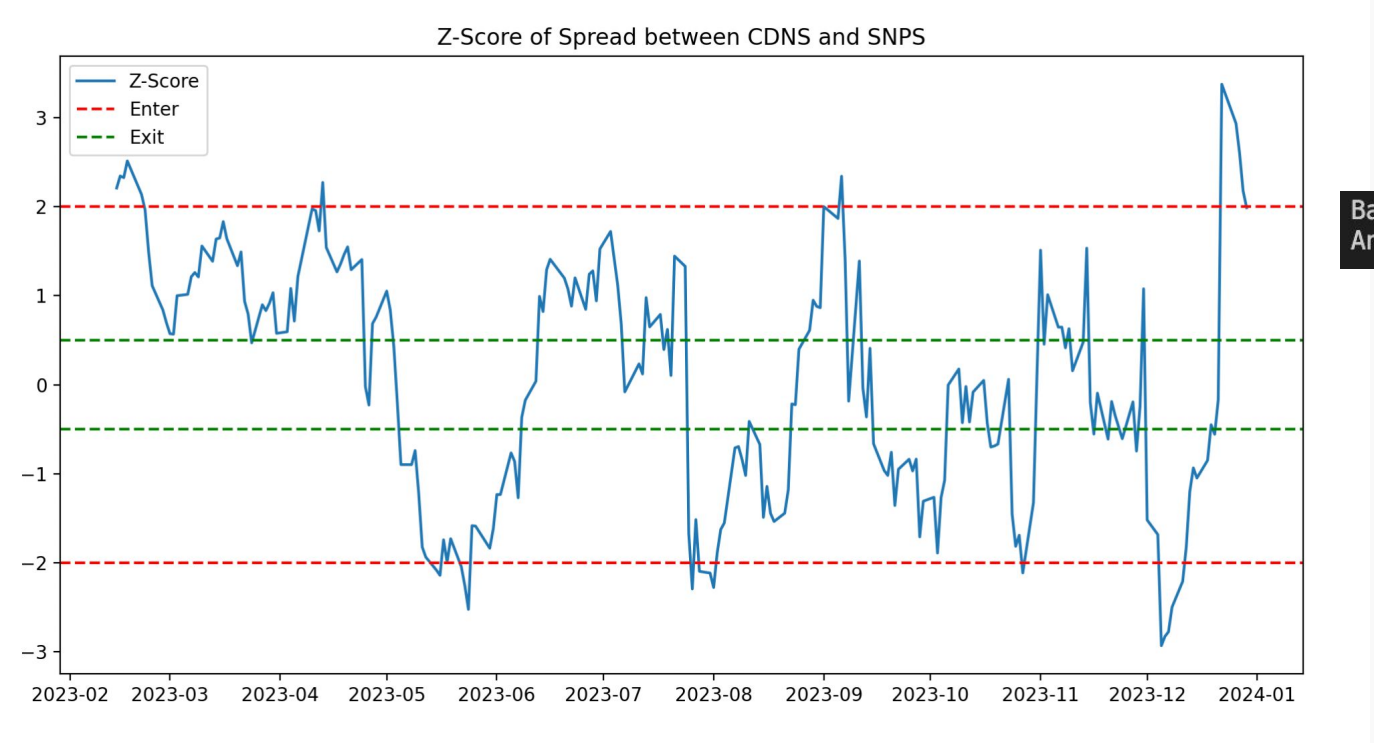
DTW Distance between ORCL and MSFT: 1.81194429983846

```
Backtesting pair: ORCL and MSFT
Annualized Return: 1.28%
```

Severely Underperforms compared to S and P

```
Backtesting pair: MSFT and MPWR
Annualized Return: -0.81%

Backtesting pair: MSFT and AVGO
Annualized Return: 1.46%

Backtesting pair: CDW and AMAT
Annualized Return: 0.62%

Backtesting pair: CDW and KLAC
Annualized Return: -0.34%

Backtesting pair: CDW and LRCX
Annualized Return: 2.36%

Backtesting pair: CDW and TER
Annualized Return: 0.64%
```

One of the closest pairs with just stocks



DTW Distance between CDNS and SNPS: 2.41194429983846

```
Backtesting pair: CDNS and SNPS
Annualized Return: 0.36%
```

Severely Underperforms compared to S and P 500

FQE Theme Colors

Text/Background – Dark 1: #333333 Dark Grey	Text/Background – Light 1: #F9F9F9 Off-White	Text/Background – Dark 2: #4F4F4F Deep Charcoal	Text/Background – Light 2: #D6C6A1 Soft Beige
Accent 1: #369347 FQE Green	Accent 2: #4682B4 Steel Blue	Accent 3: #9E7A8E Dusty Mauve	Accent 4: #585858 Charcoal Grey
Accent 5: #F1D14D Golden Yellow	Accent 6: #A3D9B1 Muted Teal	Hyperlink: #007BFF Bright Blue	Followed Hyperlink: #5A9BCF Muted Blue